AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 Hours

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Instructions:

- Total Marks: **50**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple Choice Questions: [10 marks]

Question 1: Choose the most correct answer for each of the following:

1. The critical section is controlled by the:
   a) Remainder section
   b) Entry section
   c) Entry and Exit sections
   d) Exit section
2. In scheduling, which of the following algorithms is preemptive?
   a) FCFS
   b) RR
   c) SRTF
   d) Both b and c.
3. The correct operations order for protecting a critical section using **Semaphore** is:
   a) signal() followed by wait()
   b) release() followed by acquire()
   c) wait() followed by signal()
   d) acquire() followed by release()
4. What is the primary purpose of demand paging?
   a) To reduce the time a program takes to execute.
   b) To load all pages of a process into memory at once.
   c) To load pages only when they are needed.
   d) To manage CPU scheduling.
5. Which section of code should be synchronized to prevent race conditions, in a multi-threaded environment?
   a) Thread creation
   b) Critical section
   c) Non-critical section
   d) Thread termination
6. In Priority Scheduling, a higher numbered priority typically indicates:
   a) Higher priority
   b) No specific priority level
   c) Lower priority
   d) Both choices, b and c.
7. If a process is spending more time paging than executing, then this is called:
   a) Compaction
   b) Paging
   c) Thrashing.
   d) None of the choices.
8. If a system is in the unsafe mode, this means that we have:
   a) Deadlock
   b) A possibility of a deadlock
   c) No deadlock
   d) Either a or b
9. Which of the following is an example of an open-source operating system?
   a) Windows 11
   b) Mac OS
   c) Linux
   d) Google Drive.
10. In paging, a **logical address** consists of:
    a) A segment number and an offset
    b) A base register value and a displacement
    c) A page number and a page offset
    d) A frame number and a segment limit.

Part 2: Short and Essay questions: [15 marks]

Question 2: [3 marks]

Multiprogramming offers several advantages, including increased CPU utilization and improved memory utilization. Briefly explain how multiprogramming achieves each of these benefits.

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Question 3: [4 marks]
Differentiate between Protection and Security.

Question 4: [4 marks]
Write the two methods that implement the simple **semaphore** wait() and signal() operations on global variable **S**.

Question 5: [4 marks]
Define **Linker** and **Loader** in Computer Science context.

Part 3: Problem Solving questions: [25 marks]

Question 6: [6 marks]
a) What is **Mutual Exclusion** in the context of the Critical-Section Problem?
b) What is meant by **Progress** in this context?

Question 7: [6 marks]
Consider the following page reference string: 7,5,6,3,7,3,6,5,2,8

Ref String	7	5	6	3	7	3	6	5	2	8
f1										
f2										
f3										
H/PF										

Assume that there are **three frames**, and the **Optimal algorithm** was used.
Answer the following questions:

a- Fill in the table above after copying it to your answer booklet. (5 marks)
b- Find the number of Page Faults (PF). (1 mark)

Instructions:

- in the last row, you should indicate whether we have a hit (**H**) or a page fault (**PF**)
- In the last column(s), if you do not have future pages, use **FIFO**.

Question 8: [6 marks]
Consider the following segment table:

Segment	Base	Limit
0	100	150
1	400	135
2	50	40
3	600	180
4	790	190

Find the physical addresses for the following logical addresses and deduce whether this address will be of legal reference or not. (Show the calculations)

a) Segment: 0, Logical address: 100 (2 marks)
b) Segment: 1, Logical address: 135 (2 marks)
c) Segment: 3, Logical address: 170 (2 marks)

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Question 9: [7 marks]

Consider the memory partitions of 400 KB, 100 KB, 600 KB, 300 and 200 KB, and processes of 80 KB, 250 KB, 500 KB, and 250 KB (arrived in order).

a) How would each of the first-fit, best-fit and worst-fit algorithms place the above processes in the given memory partitions. (6 marks)

b) Which algorithm makes the most efficient use of memory? (1 marks)

N.B: Use the table below and fill it in for each algorithm.

Memory Partition	Process
400	
100	
600	
300	
200	

End of questions

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